YEAR 10 SCIENCE (ADJUSTED)
PHYSICAL SCIENCES
REVISION - TEST 2

1. A dragster at Kwinana Motorplex uniformly accelerates from rest and covers the 400 m track in 7.101 s with a final speed of 85.8 m/s. Calculate the acceleration of the vehicle.

(Use $a = \frac{v - u}{t}$)

$v = 85.8$ m/s
$u = 0$ m/s
$a = ?$
$t = 7.101$ s

$a = \frac{v-u}{t}$
$= \frac{85.8 - 0}{7.101}$
$= 12.1$ m/s^2 forwards

2. The highest "ridiculous" basketball shot is currently 201.4 m from the top of Maletsuyane Falls in Lesotho. Assuming the initial vertical velocity is zero when it is released and it takes 6.41 s to fall, how fast was the ball moving vertically when it reached the ring? The acceleration due to gravity is 9.80 m/s^2. (Use $v = u + at$)

$v = ?$
$u = 0$ m/s
$a = 9.80$ m/s^2
$t = 6.41$ s

$v = u + at$
$= 0 + (9.80)(6.41)$
$= 62.8$ m/s down

3. An ES-3A Shadow jet moving at 66.7 m/s relative to an aircraft carrier lands on its flight deck and the tail hook snags the arresting wire, which stops the jet in 2.30 s. Determine the deceleration experienced by the pilot. (Use $a = \frac{v - u}{t}$)

$v = 0$ m/s
$u = 66.7$ m/s
$a = ?$
$t = 2.30$ s

$a = \frac{v-u}{t}$
$= \frac{0 - 66.7}{2.30}$
$= -29$ m/s^2 backwards

4. Complete these statements about Newton's three laws of motion.

(a) An object remains at rest, or will not change its speed or direction, unless ...

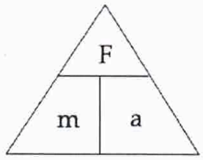
acted on by an external, unbalanced force.

(b) When an unbalanced force acts on an object, the mass of an object affects ...

the size of the acceleration or deceleration experienced.

(c) For every action, there is ...

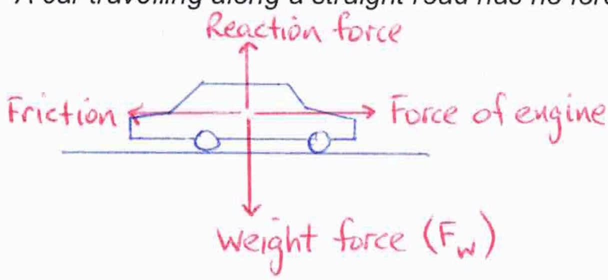
an equal and opposite reaction.



5. Complete the following table.

Force	**Mass**	**Acceleration**
100 N	20 kg	5 m/s^2
1·2 N	0.300 kg	4 m/s^2
1.2 N	2 kg	0·6 m/s^2

6. Explain why the following statement is ***false***:
A car travelling along a straight road has no forces acting on it.

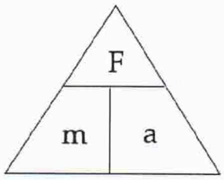


- There are 4 forces acting at all times on the car.

7. Explain in terms of Newton's First Law of Motion and ***inertia*** why it is dangerous to have loose objects inside a moving car.

- When the car brakes suddenly or has an accident, loose objects have inertia.
- They will continue to move forward and could injure the passengers.

8. Use Newton's Second Law to calculate answers for the following.



(a) A 1400 kg car accelerates at 3.0 m/s^2. What size force is needed to cause this acceleration?

$F = ?$

$m = 1400$ kg

$a = 3{\cdot}0$ m/s^2

$F = ma$

$= 1400 \times 3{\cdot}0$

$= \underline{4200 \text{ N forwards}}$

(b) A force of 160 N causes an object to accelerate at 2.0 m/s^2. What is the object's mass?

$F = 160$ N

$m = ?$

$a = 2{\cdot}0$ m/s^2

$F = ma$

$\Rightarrow m = \frac{F}{a}$

$= \frac{160}{2{\cdot}0}$

$= \underline{80 \text{ kg}}$

(c) A force of 210 N acts on a mass of 70 kg. What is the acceleration?

$F = 210$ N

$m = 70$ kg

$a = ?$

$F = ma$

$\Rightarrow a = \frac{F}{m}$

$= \frac{210}{70}$

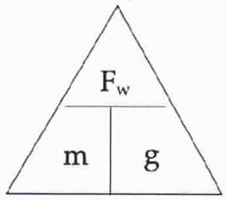
$= \underline{3{\cdot}0 \text{ m/s}^2 \text{ forwards}}$

9. (a) Explain the difference between ***mass*** and ***weight***. Give an example to help your explanation.

<u>Mass</u> - a measure of the amount of matter in an object.

<u>Weight</u> - a measure of the force due to gravity.

(b) The acceleration due to gravity on Jupiter is 22.5 m/s^2. Calculate the weight of a 450 kg object on Jupiter.



$F_w = ?$

$m = 450$ kg

$g = 22.5\ m/s^2$

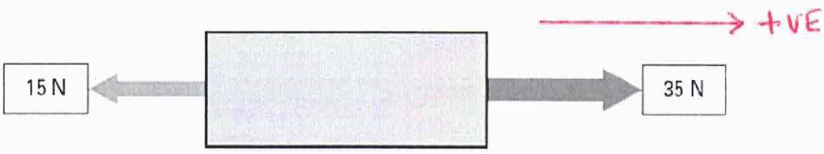
$F_w = mg$

$= 450 \times 22.5$

$= \underline{10{,}130\ N}$

10. Consider the forces acting on a body of mass 40 kg resting on a frictionless surface.

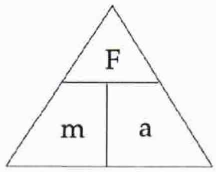
(a) Calculate the net force acting on the body.



$F = 35 - 15$

$= \underline{20N \text{ to the right}}$

(b) Calculate the acceleration of the body.



$F = 20N$

$m = 40$ kg

$a = ?$

$F = ma$

$\Rightarrow a = \frac{F}{m}$

$= \frac{20}{40}$

$= \underline{0.5\ m/s^2 \text{ to the right}}$

11. What is the difference between a ***primary safety feature*** and a ***secondary safety feature***?

Primary feature - helps prevent an accident happening.

Secondary feature - helps prevent or reduce injury to the passengers during an accident.

12. By increasing the time over which a collision occurs, the force of the impact on a car and its passengers can be decreased.

List and describe at least ***two safety features*** of a car that decrease the force of impact in this way.

Air bags - collapse in a controlled way to decelerate a passenger over a longer time.

Crumple zones - collapse in a controlled way to spread energy away from the passengers and decelerate the car more slowly.

Padded arm rests and dashboards - soft surfaces so impacts occur over a longer time, reducing the force experienced.